Muhammad Ariff Bin Abdul Razak

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Summary

Mechatronics Engineering student with 3+ years of hands-on experience in robotics software, machine learning, and autonomous systems. Proficient in ROS, C++, Python, deep learning (TensorFlow, PyTorch), SLAM, and LiDAR-based perception. Experienced in both fullstack software development and ML model research across academic and industry settings. Targeting roles in robotics software engineering, autonomous systems, or applied ML.

Education

B.Eng. Mechatronics Systems August 2022 - May 2027

Digipen Institute of Technology, Singapore Institute of Technology

- **Relevant Coursework**: Mechanics, Control Systems, Robotics, Embedded Systems
- **Methodologies**: Trained in **Agile** and **Scrum** methodologies, with hands-on experience in team collaboration, sprint planning, and iterative development for engineering projects.

Specialist Diploma in Robotics Engineering September 2022 - September 2023

Ngee Ann Polytechnic, School of Engineering (SOE)

- **Coursework**: Computer Vision, Deep Learning for Robotics
- **Technical Skills**: Hands-on experience with Keras and TensorFlow for model training and deployment, covering both supervised and unsupervised learning.
- **Areas of Interest**: Convolutional Neural Networks (CNN), Deep Neural Networks (DNN)
- Implemented waypoint navigation and data capture in the project.

Experience

KLASS Engineering Solutions

Architected and deployed scalable fullstack features for internal AI platforms and spearheaded the evaluation of state-of-the-art machine learning models to accelerate R&D pipelines.

Machine Learning Software Engineer January 2026 - August 2026

- Machine Learning Research: Benchmarked and reproduced 5+ state-of-the-art ML models, optimizing inference performance and establishing baseline metrics for future model iterations.

Fullstack Software Engineer September 2025 - January 2026

- Fullstack Feature Development: Developed and refined full-stack features for internal software tools, focusing on improving user experience, centralizing backend configurations, and ensuring system maintainability.

Singapore Institute of Technology

Worked as a student research engineer under the Student Work Scheme under a Professor to conduct research and tests

Student Research Engineer August 2024 - Present

- Designed and built a testing rig for an academic research project.

Singapore University of Technology & Design (SUTD)

Worked in SUTD's Robotics and Automation Research (ROAR) lab in a team to conduct projects,research and experiments in robotics.

Volunteer / Full Time Research Assistant August 2022 - May 2024

- Actively participated in algorithm and software design for a T1 autonomous robot that utilises 3D Lidars (Ouster).
- Prototyped a stair-climbing robot from concept to deployment, utilizing Embedded C++ for low-level motor control and SolidWorks for complex 3D modeling and mechanical integration.

Nullspace Group

Part Time / Full Time Technology Instructor November 2016 - Present

- Instructed students in STEM-focused courses at schools and learning centers, covering programming (C++, Python), robotics, and microcontrollers.
- **Certification**: MOE Instructor Certification (Oct 2018 – Oct 2026) – Authorized to teach in MOE schools and certified to instruct students in the Gifted Education Program (GEP).

Skills

- **Programming Languages**: C++, Python, Java, Embedded C
- **Machine Learning & Data Science**: Keras, TensorFlow, Pytorch , supervised and unsupervised learning
- **Robotics**: ROS, SLAM (3D LiDAR), control systems (PID), computer vision.
- **Tools** : SolidWorks, Autodesk Inventor,
- **Software:** Git, Dev-Containers , Docker , RestAPIs, Websocket , Zustand , OpenAPi , UV, Slurm.

Volunteering

- **ROSCON 2025 SINGAPORE(Intrinsic , Open Robotics):** Supported event operations , contributed to attendee experience, workshops and logistics